

MA 611: Homework #1

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1 Finite dimensional normed linear space

Question: Let Y be a finite dimensional linear subspace of a normed linear space X , and let $x_0 \in X$, $x_0 \notin Y$. Then there exist a point $y_0 \in Y$ such that

$$\inf_{y \in Y} \|x_0 - y\| = \|x_0 - y_0\|.$$

Proof: Define the distance between x_0 and Y as

$$d = \inf_{y \in Y} \|x_0 - y\|.$$

So we can find a sequence $\{y_n\}_{n=1}^{\infty}$ such that

$$\|y_n - x_0\| \leq d + \frac{1}{n}, \quad \forall n > N.$$

Since $\{y_n\}_{n=1}^{\infty}$ is a bounded sequence, it has a convergent subsequence $\{y_{n_k}\}_{k=1}^{\infty}$, and let

$$y_0 = \lim_{k \rightarrow \infty} y_{n_k}.$$

Since Y is a finite dimensional linear subspace, by Theorem 4.3.2 in Friedman's book, we know that Y is closed and $y_0 \in Y$.

We need to show that $\|x_0 - y_0\| = d$.

On one hand, it is obvious that

$$\|x_0 - y_0\| \geq d.$$

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On the other hand,

$$\|x_0 - y_0\| \leq \|y_0 - y_{n_k}\| + \|y_{n_k} - x_0\| \rightarrow d, \quad \text{as } k \rightarrow \infty.$$

So y_0 is just what we want to find.

2 Linear transformation

Question: Find the norm of the operator $A \in \mathcal{B}(X)$ given by $(Af)(t) = tf(t)$, $0 \leq t \leq 1$, where

- (a) $X = C([0, 1])$
- (b) $X = L^p([0, 1])$, $1 \leq p < \infty$.

Solution:

- (a) On one hand,

$$\|Af\| = \|tf(t)\| \leq \|t\| \|f(t)\| = \|f(t)\|, \quad \forall f \in C([0, 1]),$$

so $\|A\| \leq 1$.

On the other hand, let $f \equiv 1$, then $\|f\| = 1$ and

$$\|A\| \geq \|tf(t)\| = 1.$$

So $\|A\| = 1$.

- (b) On one hand,

$$\|Af\|_p = \|tf(t)\|_p \leq \|t\|_p \|f(t)\|_p = \left(\frac{1}{p+1}\right)^{\frac{1}{p}} \|f(t)\|_p, \quad \forall f \in L^p([0, 1]),$$

so $\|A\| \leq \left(\frac{1}{p+1}\right)^{\frac{1}{p}}$.

On the other hand, let $f \equiv 1$, then $\|f\|_p = 1$ and

$$\|A\| \geq \|tf(t)\|_p = \left(\frac{1}{p+1}\right)^{\frac{1}{p}}.$$

So $\|A\| = \left(\frac{1}{p+1}\right)^{\frac{1}{p}}$.

3 Linear transformation

Question: A linear operator from a normed linear space X into a normed linear space is bounded if and only if it maps bounded sets onto bounded sets.

Proof: $\Rightarrow$): Suppose $T : X \rightarrow Y$ is a bounded linear operator, then

$$\|Tx\| \leq \|T\|\|x\|,$$

which means that if x is bounded then Tx should also be bounded.

$\Leftarrow$): Choose the closed unit ball $B = \overline{B_1(0)} \subset X$, then B is bounded. so

$$\alpha = \sup_{x \in B} \|Tx\| < \infty.$$

Since $\forall x \in X, \frac{x}{\|x\|} \in B$, so

$$\begin{aligned} & \left\| T \left(\frac{x}{\|x\|} \right) \right\| \leq \alpha \\ \Rightarrow & \|Tx\| \leq \alpha\|x\|. \end{aligned}$$

It implies that T is bounded.

MA 611: Homework #2

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1 Problem 3.1.5

Question: Prove that the spaces l^1, l^∞, s, c, c_0 and $C[a, b]$ are complete metric spaces.

Proof:

1.1 l^1

Let $\{x_n\}_{n=1}^\infty \subset l^1$ be a Cauchy sequence, where $x_n = (\xi_1^{(n)}, \dots, \xi_k^{(n)}, \dots)$, $\sum_{k=1}^\infty |\xi_k^{(n)}| < \infty$.

$\forall \varepsilon > 0, \exists N$ such that $\forall m, n > N$,

$$\|x_n - x_m\|_{l^1} = \sum_{k=1}^\infty |\xi_k^{(n)} - \xi_k^{(m)}| < \varepsilon. \quad (1.1)$$

It implies that for fixed k , $\{\xi_k^{(n)}\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{R}^1$. So we can denote $\lim_{n \rightarrow \infty} \xi_k^{(n)} = \xi_k^{(0)}$ and let $x_0 = (\xi_1^{(0)}, \dots, \xi_k^{(0)}, \dots)$.

In Eq.(1.1), let $m \rightarrow \infty$, we can get

$$\|x_n - x_0\|_{l^1} \leq \varepsilon, \quad \text{for } n > N, \quad (1.2)$$

which means $\lim_{n \rightarrow \infty} x_n = x_0$ in the sense of l^1 .

Besides, since

$$\|x_0\|_{l^1} \leq \|x_0 - x_n\|_{l^1} + \|x_n\|_{l^1} < \infty,$$

we know that $x_0 \in l^1$ and l^1 is complete.

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1.2 l^∞

Let $\{x_n\}_{n=1}^\infty \subset l^\infty$ be a Cauchy sequence, where $x_n = (\xi_1^{(n)}, \dots, \xi_k^{(n)}, \dots)$, $\sup_k |\xi_k^{(n)}| < \infty$.

$\forall \varepsilon > 0$, $\exists N$ such that $\forall m, n > N$,

$$\|x_n - x_m\|_{l^\infty} = \sup_k |\xi_k^{(n)} - \xi_k^{(m)}| < \varepsilon. \quad (1.3)$$

It implies that for fixed k , $\{\xi_k^{(n)}\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{R}^1$. So we can denote $\lim_{n \rightarrow \infty} \xi_k^{(n)} = \xi_k^{(0)}$ and let $x_0 = (\xi_1^{(0)}, \dots, \xi_k^{(0)}, \dots)$.

In Eq.(1.3), let $m \rightarrow \infty$, we can get

$$\|x_n - x_0\|_{l^\infty} \leq \varepsilon, \quad \text{for } n > N, \quad (1.4)$$

which means $\lim_{n \rightarrow \infty} x_n = x_0$ in the sense of l^∞ .

Besides, since

$$\|x_0\|_{l^\infty} \leq \|x_0 - x_n\|_{l^\infty} + \|x_n\|_{l^\infty} < \infty,$$

we know that $x_0 \in l^\infty$ and l^∞ is complete.

1.3 s

Let $\{x_n\}_{n=1}^\infty \subset s$ be a Cauchy sequence, where $x_n = (\xi_1^{(n)}, \dots, \xi_k^{(n)}, \dots)$ with

$$\sum_{k=1}^\infty \frac{1}{2^k} \frac{|\xi_k^{(n)}|}{1 + |\xi_k^{(n)}|} < \infty.$$

$\forall \varepsilon > 0$,

$$\|x_n - x_m\|_s = \sum_{k=1}^\infty \frac{1}{2^k} \frac{|\xi_k^{(n)} - \xi_k^{(m)}|}{1 + |\xi_k^{(n)} - \xi_k^{(m)}|} \rightarrow 0, \quad \text{as } n, m \rightarrow \infty. \quad (1.5)$$

So $|\xi_k^{(n)} - \xi_k^{(m)}| \rightarrow 0$ as $n, m \rightarrow \infty$.

It implies that for fixed k , $\{\xi_k^{(n)}\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{R}^1$. So we can denote $\lim_{n \rightarrow \infty} \xi_k^{(n)} = \xi_k^{(0)}$ and let $x_0 = (\xi_1^{(0)}, \dots, \xi_k^{(0)}, \dots)$.

On one hand, $\forall \varepsilon > 0$, $\exists K$ such that

$$\sum_{k > K} \frac{1}{2^k} < \frac{\varepsilon}{2}$$

On the other hand, $\exists N$ such that $\forall n > N, \forall k \leq K$,

$$|\xi_k^{(n)} - \xi_k^{(0)}| < \frac{\varepsilon}{2}.$$

Hence, we have

$$\begin{aligned} \|x_0 - x_n\|_s &= \sum_{k=1}^{\infty} \frac{1}{2^k} \frac{|\xi_k^{(n)} - \xi_k^{(0)}|}{1 + |\xi_k^{(n)} - \xi_k^{(0)}|} \\ &= \sum_{k=1}^K \frac{1}{2^k} \frac{|\xi_k^{(n)} - \xi_k^{(0)}|}{1 + |\xi_k^{(n)} - \xi_k^{(0)}|} + \sum_{k=K+1}^{\infty} \frac{1}{2^k} \frac{|\xi_k^{(n)} - \xi_k^{(0)}|}{1 + |\xi_k^{(n)} - \xi_k^{(0)}|} \\ &\leq \sum_{k=1}^K \frac{1}{2^k} |\xi_k^{(n)} - \xi_k^{(0)}| + \sum_{k=K+1}^{\infty} \frac{1}{2^k} \\ &< \frac{\varepsilon}{2} \sum_{k=1}^K \frac{1}{2^k} + \frac{\varepsilon}{2} \\ &< \varepsilon, \end{aligned}$$

which means $\lim_{n \rightarrow \infty} x_n = x_0$ in the sense of s .

Besides, since

$$\|x_0\|_s \leq \|x_0 - x_n\|_s + \|x_n\|_s < \infty,$$

we know that $x_0 \in s$ and s is complete.

1.4 c_0

Let $\{x_n\}_{n=1}^{\infty} \subset c_0$ be a Cauchy sequence, where $x_n = (\xi_1^{(n)}, \dots, \xi_k^{(n)}, \dots)$ with $\sup_k |\xi_k^{(n)}| < \infty$

and $\lim_{k \rightarrow \infty} \xi_k^{(n)} = 0$.

Since

$$\|x_n - x_m\|_{l^\infty} = \sup_k |\xi_k^{(n)} - \xi_k^{(m)}| > |\xi_k^{(n)} - \xi_k^{(m)}|, \quad (1.6)$$

for fixed k , $\{\xi_k^{(n)}\}_{n=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}^1$. So we can denote $\lim_{n \rightarrow \infty} \xi_k^{(n)} = \xi_k^{(0)}$ and let $x_0 = (\xi_1^{(0)}, \dots, \xi_k^{(0)}, \dots)$.

We need to prove $x_0 \in c_0$.

$\forall \varepsilon > 0$, $\exists N$ such that

$$|\xi_k^{(n)} - \xi_k^{(N)}| < \frac{\varepsilon}{2}, \quad \forall n > N.$$

Let $n \rightarrow \infty$, we can get

$$|\xi_k^{(0)} - \xi_k^{(N)}| \leq \frac{\varepsilon}{2}.$$

Since $\lim_{k \rightarrow \infty} \xi_k^{(N)} = 0$, there exists $K > 0$ such that $\forall k > K$, $|\xi_k^{(N)}| < \frac{\varepsilon}{2}$.

So for $k > K$,

$$|\xi_k^0| \leq |\xi_k^0 - \xi_k^N| + |\xi_k^N| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon.$$

It implies that $\{\xi_k^0\}_{k=1}^\infty$ is a sequence convergent to 0, which means $x_0 \in c_0$.

We also need to prove that $\lim_{n \rightarrow \infty} x_n = x_0$.

$\forall \varepsilon > 0$, $\exists N$ such that $\forall m, n > N$,

$$\|x_n - x_m\|_{l^\infty} = \sup_k |\xi_k^{(n)} - \xi_k^{(m)}| < \varepsilon. \quad (1.7)$$

Let $m \rightarrow \infty$, we can get

$$\|x_n - x_0\|_{l^\infty} \leq \varepsilon, \quad \text{for } n > N, \quad (1.8)$$

which means $\lim_{n \rightarrow \infty} x_n = x_0$ in the sense of l^∞ .

1.5 c

Let $\{x_n\}_{n=1}^\infty \subset c$ be a Cauchy sequence, where $x_n = (\xi_1^{(n)}, \dots, \xi_k^{(n)}, \dots)$ with $\sup_k |\xi_k^{(n)}| < \infty$ and

$$\lim_{k \rightarrow \infty} \xi_k^{(n)} < \infty, \quad \forall n \in N.$$

Given $\varepsilon > 0$, $\exists N$, such that $\forall m, n > N$,

$$\sup_k |\xi_k^n - \xi_k^m| < \varepsilon. \quad (1.9)$$

So for fixed k , $\{\xi_k^{(n)}\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{R}^1$. Then we can denote $\lim_{n \rightarrow \infty} \xi_k^{(n)} = \xi_k^{(0)}$ and let $x_0 = (\xi_1^{(0)}, \dots, \xi_k^{(0)}, \dots)$.

Let $m \rightarrow \infty$ in (1.9), we know that

$$\sup_k |\xi_k^n - \xi_k^0| < \varepsilon,$$

which means $x_n \rightarrow x_0$ as $n \rightarrow \infty$.

We need to prove $x_0 \in c$, that is to say, $\lim_{k \rightarrow \infty} \xi_k^{(0)}$ exists.

Notice the fact that $\exists K$, such that $\forall p, q > K$,

$$|\xi_p^n - \xi_q^n| < \varepsilon.$$

Then for $p, q > K$ and $n > N$, we have

$$\begin{aligned} \|\xi_p^0 - \xi_q^0\|_{l^\infty} &\leq \|\xi_p^0 - \xi_p^n\|_{l^\infty} + \|\xi_p^n - \xi_q^n\|_{l^\infty} + \|\xi_q^n - \xi_q^0\|_{l^\infty} \\ &\leq 3\varepsilon, \end{aligned}$$

which means $x_0 \in c$.

1.6 $C[a, b]$

Let $\{x_n(t)\}_{n=1}^\infty$ be a Cauchy sequence in $C[a, b]$. Then $\forall \varepsilon > 0, \exists N$ such that

$$\max_{t \in [a, b]} |x_n(t) - x_m(t)| < \varepsilon, \quad \forall n, m > N. \quad (1.10)$$

For fixed $t_0 \in [a, b]$, $\{x_n(t)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{R}$. So we can define the function $x_0(t) = \lim_{n \rightarrow \infty} x_n(t)$. By (1.10) we know that the convergence of $\{x_n(t)\}_{n=1}^\infty$ is uniformly convergent on $[a, b]$.

Hence, $\forall \varepsilon > 0, \exists N$ such that

$$|x_N(t) - x_0(t)| < \varepsilon, \quad \forall t \in [a, b]. \quad (1.11)$$

Since $x_N(t) \in C[a, b]$, for $t_0 \in [a, b]$, there exists a $\delta > 0$ such that $\forall t \in (t_0 - \delta, t_0 + \delta)$, we have

$$|x_N(t) - x_N(t_0)| < \varepsilon. \quad (1.12)$$

By (1.11) and (1.12), we know that

$$\begin{aligned} |x_0(t) - x_0(t_0)| &\leq |x_0(t) - x_N(t)| + |x_N(t) - x_N(t_0)| + |x_N(t_0) - x_0(t_0)| \\ &\leq 3\varepsilon, \end{aligned}$$

which means $x_0(t) \in C[a, b]$.

2 Problem 3.16

Question: Prove that the spaces l^1, s, c, c_0 are separable metric spaces.

Proof:

2.1 l^1

The set $P = \{y = (\eta_1, \eta_2, \dots, \eta_n, 0, 0, \dots) | \eta_k \in \mathbb{Q}, 1 \leq k \leq n\}$ is a countable subset of l^1 .

Given $\varepsilon > 0$ and $x = (\xi_1, \dots, \xi_k, \dots) \in l^1$, then $\exists N$ such that

$$\sum_{k=N+1}^{\infty} |\xi_k| < \frac{\varepsilon}{2}.$$

$\mathbb{Q}$ is dense in $\mathbb{R}$, so there is a $y \in P$ such that

$$\sum_{k=1}^N |\xi_k - \eta_k| < \frac{\varepsilon}{2}.$$

Hence,

$$\|x - y\|_{l^1} = \sum_{k=1}^N |\xi_k - \eta_k| + \sum_{k=N+1}^{\infty} |\xi_k| < \varepsilon.$$

It implies that $\overline{P} = l^1$.

2.2 s

Also consider the set $P = \{y = (\eta_1, \eta_2, \dots, \eta_n, 0, 0, \dots) | \eta_k \in \mathbb{Q}, 1 \leq k \leq n\}$ is a countable subset of s .

For $x = (\xi_1, \dots, \xi_k, \dots) \in s$, since $\mathbb{Q}$ is dense in $\mathbb{R}$ and the function $\varphi(t) = \frac{t}{1+t}$ is increasing, $\forall \varepsilon > 0$, there is a $y \in P$ such that

$$\sum_{k=1}^N \frac{1}{2^k} \frac{|\xi_k - \eta_k|}{1 + |\xi_k - \eta_k|} < \frac{\varepsilon}{2}.$$

Similarly as Section 2.1, we know that $\overline{P} = s$.

2.3 c_0

Also, since the set $P = \{y = (\eta_1, \eta_2, \dots, \eta_n, 0, 0, \dots) | \eta_k \in \mathbb{Q}, 1 \leq k \leq n\}$ is a countable subset of c_0 .

Given $\varepsilon > 0$ and $x = (\xi_1, \dots, \xi_k, \dots) \in l^1$, then $\exists N$ such that

$$\sup_{k \geq N+1} |\xi_k| < \frac{\varepsilon}{2}.$$

$\mathbb{Q}$ is dense in $\mathbb{R}$, so there is a $y \in P$ such that

$$\sup_{1 \leq k \leq N} |\xi_k - \eta_k| < \frac{\varepsilon}{2}.$$

Hence,

$$\|x - y\|_{l^\infty} \leq \sup_{1 \leq k \leq N} |\xi_k - \eta_k| + \sup_{k \geq N+1} |\xi_k| < \varepsilon.$$

It implies that $\overline{P} = c_0$. Hence, c_0 is separable.

2.4 c

Consider the set $P1 = \{y = (\eta_1, \eta_2, \dots, \eta_n, \eta_n, \eta_n, \dots) | \eta_k \in \mathbb{Q}, 1 \leq k \leq n\}$, which is countable.

Then $\forall y = (\eta_1, \eta_2, \dots, \eta_n, \eta_n, \eta_n, \dots) \in P1$, $\lim_{k \rightarrow \infty} \eta_k = \eta_n$, which means $P1 \subset c$.

Following the way in Section 2.3, it is easy to know that $\overline{P1} = c$.

3 Problem 3.1.7

Question: Prove that the space l^∞ is not separable.

Proof:

Consider the set of all sequences made of elements 0, 1 :

$$S = \{x = (\xi_1, \dots, \xi_k, \dots) \in l^\infty | \xi_k = 0 \text{ or } 1, \text{ for } k = 1, 2, \dots\}.$$

It is obvious that $S \subset l^\infty$.

$\forall x \in S$ can be seen as a binary representation of a number $\alpha = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k} \in [0, 1]$. Since the interval $[0, 1]$ is uncountable, S is also uncountable. Besides, if $x, y \in S$ and $x \neq y$, then $\|x - y\|_{l^\infty} = 1$, which means there are uncountable many sequences of zeros and ones.

Consider the set made of the balls centered at $x \in S$ with radius $\frac{1}{3}$:

$$B = \{B_{1/3}(x) | x \in S, B_{1/3}(x) \cap B_{1/3}(y) = \emptyset, \text{ if } x \neq y\}.$$

It is obvious that B is also uncountable.

Let M be any dense subset of l^∞ , then $\forall B_{1/3}(x) \in B$ must contain a element of M . It implies that B should be a uncountable set. Hence, l^∞ is not separable.

4 Problem 3.1.8

Question: Construct, in each of the spaces $l^1, l^\infty, s, c, c_0, C[a, b]$, bounded closed set that are not sequentially compact.

Solution:

4.1 l^1, l^∞, c, c_0

Consider the sequence $X = \{x_n\}_{n=1}^\infty$. $\forall x_n = (\xi_1^{(n)}, \dots, \xi_k^{(n)}, \dots) \in X$ is defined as

$$\xi_k^{(n)} = \begin{cases} 1, & k = n, \\ 0, & k \neq n. \end{cases}$$

It is obvious that

$$\|x_n\|_{l^1} = \|x_n\|_{l^\infty} = 1.$$

So X is bounded.

Besides, for fixed n , $\lim_{k \rightarrow \infty} \xi_k^{(n)} = 0$. So $X \subset c_0 \subset c \subset l^\infty$. And it is obvious that $X \subset l^1$.

However, X has no convergent subsequence.

4.2 s

Consider the set $X = \{x_n = (n, 0, 0, \dots)\}$. Then $\forall x_n \in X$,

$$\|x_n\|_s = \frac{1}{2} \cdot \frac{n}{1+n} < \frac{1}{2},$$

which means X is bounded.

However,

$$\|x_{n+1} - x_n\|_s = \frac{1}{4},$$

which means X does not have convergent subsequence.

4.3 $C[a, b]$

Without of generality, we can assume $a = 0, b = 1$, that is to say, we consider the space $C[0, 1]$.

Construct the sequence $\{f_n\}_{n=1}^\infty$ as

$$f_n(x) = \begin{cases} 1 - nx, & x \in [0, \frac{1}{n}], \\ 0, & x \in (\frac{1}{n}, 1]. \end{cases}$$

Then $\|f_n\|_\infty = 1$ so they are bounded. However,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \in (0, 1]. \end{cases}$$

It is obvious that $f(x) \notin C[0, 1]$.

MA 611: Homework #3

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1 Problem 3.5.4

Question: A subset F of a compact metric space is compact $\Leftrightarrow$ it is closed.

Proof: " $\Rightarrow$ "

Let $x \in \bar{F}$, then there exists a sequence $\{x_n\} \subset F$, such that $x_n \rightarrow x$. Since the subset F is compact in a compact space, by Theorem 3.5.4 in Friedman's book, F is also sequentially compact. Hence, $x \in F$ and F is closed,

" $\Leftarrow$ " Let X be the whole compact metric space and F is the closed subset of X . For any open cover $\mathcal{F} \supset F$, we can let $\mathcal{F}_1 = \mathcal{F} \cup \{X \setminus F\}$ is an open cover of X . Since X is compact, so the open cover of X , named $\mathcal{F}_1$ has finite subcover. It is easy to know that F also has finite subcover.

2 Problem 3.5.5

Question: A subset Y of a metric space is totally bounded $\Leftrightarrow$ its closure $\bar{Y}$ is totally bounded.

Proof: " $\Rightarrow$ "

Y is totally bounded means for $\forall \varepsilon > 0$, there exists a finite number of balls $B(x_n, \varepsilon)$ with $x_n \in Y$ such that

$$\bigcup_{n=1}^N B(x_n, \varepsilon) \supset Y.$$

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It is obvious that $\bigcup_{n=1}^N \overline{B(x_n, \varepsilon)} \supset \bar{Y}$ and for each n , $B(x_n, 2\varepsilon) \supset \overline{B(x_n, \varepsilon)}$. Then

$$\bigcup_{n=1}^N B(x_n, 2\varepsilon) \supset \bigcup_{n=1}^N \overline{B(x_n, \varepsilon)} \supset Y.$$

So $\bigcup_{n=1}^N B(x_n, 2\varepsilon)$ is the cover of $\bar{Y}$ and $\bar{Y}$ is totally bounded.

” $\Leftarrow$ ”

For any $\varepsilon > 0$, suppose $\bigcup_{n=1}^N B(x_n, \varepsilon)$ be a cover of $\bar{Y}$. That is to say,

$$\bigcup_{n=1}^N B(x_n, \varepsilon) \supset \bar{Y} \supset Y.$$

For $\forall x_n \in \bar{Y}$, $\exists y_n \in Y$, such that

$$d(x_n, y_n) < \varepsilon.$$

Then, $\forall x \in B(x_n, \varepsilon)$,

$$d(x, y_n) \leq d(x, x_n) + d(x_n, y_n) < 2\varepsilon,$$

which means $B(x_n, \varepsilon) \subset B(y_n, 2\varepsilon)$.

So

$$\bigcup_{n=1}^N B(y_n, 2\varepsilon) \supset \bigcup_{n=1}^N B(x_n, \varepsilon) \supset \bar{Y} \supset Y,$$

where $y_n \in Y$.

Then $\bigcup_{n=1}^N B(y_n, 2\varepsilon)$ is a cover of Y and Y is totally bounded.

3 Problem 3.5.9

Question: Prove that a closed set K of points $x = (x_1, x_2, \dots)$ in l^p ($1 \leq p < \infty$) is compact if

- (i) $\sum_{n=1}^{\infty} |x_n|^p \leq C$ for all $x \in K$, where C is a constant,
- (ii) for any $\varepsilon > 0$, there is an n_0 such that $\sum_{n=n_0}^{\infty} |x_n|^p \leq \varepsilon$ for all $x \in K$.

Proof: By Theorem 3.5.6 in Friedman's book, we just need to show that the set K is totally bounded if K satisfies (i) and (ii).

- (i) means $\forall x \in K$, $\|x\|_p^2 < C$, which implies that K is uniformly bounded.

For any $\varepsilon > 0$, by (ii), we can find n_0 such that $\sum_{n=n_0}^{\infty} |x_n|^p \leq \frac{\varepsilon}{2}$ for all $x \in K$.

Let

$$Y_\varepsilon = \{y \in K : y = (y_1, y_2, \dots, y_{n_0}, 0, 0, \dots)\}.$$

Then it is easy to know that Y_ε is closed and compact so it has a finite cover $\cup_{k=1}^K B(y^{(k)}, \frac{\varepsilon}{2})$, where $y^{(k)} = (y_1^{(k)}, y_2^{(k)}, \dots, y_{n_0}^{(k)}, 0, 0, \dots) \in Y_\varepsilon$.

Let

$$\mathcal{B} = \cup_{k=1}^K B(y^{(k)}, \varepsilon),$$

we want to show that $\mathcal{B}$ is a cover of K .

For any $x \in K$, by the definition of Y_ε , there exists $y \in Y_\varepsilon$ such that $d(x, y) < \frac{\varepsilon}{2}$. Besides, there exists some $y^{(k)}$ such that $y \in B(y^{(k)}, \frac{\varepsilon}{2})$, i.e. $d(y, y^{(k)}) < \frac{\varepsilon}{2}$. So we have

$$d(x, y^{(k)}) \leq d(x, y) + d(y, y^{(k)}) < \varepsilon,$$

which means $x \in B(y^{(k)}, \varepsilon)$.

Thus, $\mathcal{B}$ is a cover of K and K is totally bounded and compact.

4 Problem 4.8.1

Question: Let X be a normed linear space and let $\{x_n\} \subset X$. A point y_0 is the limit of linear combination $\sum_{j=1}^n c_j x_j \Leftrightarrow x^*(y_0) = 0$ for all x^* for which $x^*(x_j) = 0$ for $1 \leq j < \infty$.

Proof: " $\Rightarrow$ "

Let $s_n = \sum_{j=1}^n c_j x_j$, then

$$y_0 = \lim_{n \rightarrow \infty} s_n.$$

Then,

$$\begin{aligned} x^*(x_j) &= 0, \quad \forall j, \\ x^*(s_n) &= \sum_{j=1}^n c_j x^*(x_j) = 0, \quad \forall n, \\ x^*(y) &= x^*\left(\lim_{n \rightarrow \infty} s_n\right) = \lim_{n \rightarrow \infty} x^*(s_n) = 0. \end{aligned}$$

" $\Leftarrow$ "

Let $Y = \text{span} \{x_1, x_2, \dots, x_n, \dots\}$ and $y_0 \in X$ satisfying that $x^*(y_0) = 0$ for all x^* for which $x^*(x_j) = 0$ for $1 \leq j < \infty$. Then for any $y \in Y$, we have $y = \sum_{j=1}^n c_j x_j$ and

$$x^*(y) = \sum_{j=1}^n c_j x^*(x_j) = 0, \tag{4.1}$$

where $x * (x_j) = 0$ for $1 \leq j < \infty$.

We claim that $d(y_0, Y) = \inf_{y \in Y} \|y - y_0\| = 0$. If not, $d(y_0, Y) > 0$, we want to get a contradiction.

By Theorem 4.8.3, there exists a point $x^* \in X^*$ such that $x^*(y_0) = 1$ and $x^*(y) = 0$ for all $y = \sum_{j=1}^n c_j x_j \in Y$, which contradicts with (4.1).

Since we have known that $d(y_0, Y) = 0$, there should exist a linear combination $\sum_{j=1}^n c_j x_j$ whose limit is y_0 .

5 Extreme points

Question: Let $c_0 = \{x = (x_1, x_2, \dots) : \lim_{n \rightarrow \infty} x_n = 0\}$ and $\|x\|_\infty = \sup_n |x_n|$. Let $E = \{x \in c_0, \|x\|_\infty \leq 1\}$, then E has no extreme points.

Proof:

Step 1. Claim that $\forall x \in E$, there exists $y, z \in E$ and $x \neq y \neq z$ such that $x = \frac{1}{2}x + \frac{1}{2}y$.

Let $x = (x_1, x_2, \dots) \in E$. Since $\lim_{n \rightarrow \infty} x_n = 0$, there exists N such that for any $n > N$, $|x_n| < \frac{1}{2}$.

Choose $y = (x_1, \dots, x_N, 0, \dots)$ and $z = (x_1, \dots, x_N, 2x_{N+1}, 2x_{N+2}, \dots)$. It is easy to know that $\|y\|_\infty \leq 1$ and $\|z\|_\infty \leq 1$, which means $y, z \in E$. Besides,

$$x = \frac{1}{2}x + \frac{1}{2}y.$$

Step 2. Suppose $\{x\}$ is the extreme point of E , then x can not be written as $x = \frac{1}{2}y + \frac{1}{2}z$ where $y, z \in E$ and $x \neq y, x \neq z$. It is contradicts with the claim in *Step 1*.

Hence, E has no extreme points.

MA 611: Homework #4

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1 Problem 4.5.2

Question: Show that

$$\int_0^{2\pi} \left| \frac{\sin(n + \frac{1}{2})x}{\sin(\frac{1}{2}x)} \right| dx \rightarrow \infty, \quad \text{as } n \rightarrow \infty \quad (1.1)$$

Proof:

$$\begin{aligned} & \int_0^{2\pi} \left| \frac{\sin(n + \frac{1}{2})x}{\sin(\frac{1}{2}x)} \right| dx \\ = & \int_0^{2\pi} \frac{|\sin(n + \frac{1}{2})x|}{\sin(\frac{1}{2}x)} dx \\ \geq & \int_0^{2\pi} \frac{|\sin(n + \frac{1}{2})x|}{\frac{1}{2}x} dx \\ = & 2 \int_0^{(2n+1)\pi} \frac{|\sin y|}{y} dy, \quad \text{where } y = \left(n + \frac{1}{2}\right)x \\ \geq & 2 \sum_{k=0}^{2n} \frac{\int_0^{\pi} \sin y dy}{k+1} \\ = & 2 \sum_{k=1}^{2n} \frac{1}{k}. \end{aligned}$$

Since $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$, we can get (1.1).

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2 Problem 4.5.3

Question: The Fourier series of an integrable function $f(x)$ defined on $(0, 2\pi)$ is the series

$$s(x) = \sum_{m=-\infty}^{\infty} a_m e^{imx},$$

where $a_m = \frac{1}{2\pi} \int_0^{2\pi} f(\xi) e^{-im\xi} d\xi$.

Set $s_n(x) = \sum_{m=-n}^n a_m e^{imx}$, $f(x) = f(x + 2\pi)$ ($0 < x < 2\pi$). Prove that

$$s_n(y) = \frac{1}{2\pi} \int_0^{2\pi} f(y+x) D_n(x) dx,$$

where

$$D_n(x) = \frac{\sin(n + \frac{1}{2})x}{\sin(\frac{1}{2}x)}.$$

Proof: It is easy to know that

$$\begin{aligned} D_n(x) &= \frac{\sin(n + \frac{1}{2})x}{\sin(\frac{1}{2}x)} \\ &= 1 + 2 \sum_{k=1}^n \cos(kx) \\ &= \sum_{k=-n}^n e^{ikx}. \end{aligned} \tag{2.1}$$

By the definition, we know that

$$\begin{aligned} s_n(y) &= \sum_{m=-n}^n a_m e^{imy} \\ &= \sum_{m=-n}^n \frac{1}{2\pi} \left(\int_0^{2\pi} f(\xi) e^{-im\xi} d\xi \right) e^{imy} \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(\xi) \left(\sum_{m=-n}^n e^{im(y-\xi)} \right) d\xi \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(y+x) \left(\sum_{m=-n}^n e^{imx} \right) dx, \quad \text{where } x = y - \xi, \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(y+x) D_n(x) dx. \end{aligned}$$

3 Problem 4.5.4

Question: Denote by X the Banach space of all continuous functions $f(x)$ on $0 \leq x \leq 2\pi$, with $f(0) = f(2\pi)$, provided with the uniform norm. Denote by Y the Euclidean space R^1 . Prove that the linear operator

$$T_n(f) = \frac{1}{2\pi} \int_0^{2\pi} f(x) D_n(x) dx$$

from X into Y is bounded, and

$$\|T_n\| = \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx. \quad (3.1)$$

Proof: On one hand, for each $f \in X$,

$$\begin{aligned} |T_n(f)| &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(x)| |D_n(x)| dx \\ &\leq \max_{x \in [0, 2\pi]} |f(x)| \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx \\ &= \|f\| \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx, \end{aligned}$$

which means

$$\|T_n\| \leq \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx. \quad (3.2)$$

On the other hand, we want to prove that $\|T_n\| \geq \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx$.

It is easy to know that

$$D_n(x) \begin{cases} = 0, & x = \delta_{n,k}, \\ \neq 0, & \text{otherwise,} \end{cases}$$

where $\delta_{n,k} = \frac{k\pi}{n+\frac{1}{2}}$.

Given $\varepsilon > 0$, for each n , construct a continuous function $\tilde{f}_n \in X$ as

$$\tilde{f}_n(x) = \begin{cases} \operatorname{sgn}(D_n(x)), & x = [0, 2\pi] \setminus \Delta, \\ \text{linear functions,} & x \in \Delta, \end{cases}$$

where $\Delta = (\cup(\delta_{n,k} - \varepsilon, \delta_{n,k} + \varepsilon)) \cap [0, 2\pi]$.

It is easy to verify that: (1) $\tilde{f}_n(x) D_n(x) \geq 0$, for $\forall x \in [0, 2\pi]$; (2) $|D_n(x)| \leq \varepsilon$ for $\forall x \in \Delta$; (3) $\|\tilde{f}_n\| = 1$.

So we have

$$\begin{aligned}
|T_n(\tilde{f}_n)| &= \frac{1}{2\pi} \int_0^{2\pi} \tilde{f}_n(x) D_n(x) dx \\
&\geq \frac{1}{2\pi} \int_{[0, 2\pi] \setminus \Delta} |D_n(x)| dx - \frac{1}{2\pi} \int_{\Delta} |D_n(x)| dx \\
&\geq \frac{1}{2\pi} \int_0^1 |D_n(x)| dx - 2 \frac{1}{2\pi} \int_{\Delta} |D_n(x)| dx \\
&\geq \frac{1}{2\pi} \int_0^1 |D_n(x)| dx - 2\varepsilon.
\end{aligned}$$

Let $\varepsilon \rightarrow 0$, we know that

$$\|T_n\| \geq \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx. \quad (3.3)$$

By (3.2) and (3.3), we can conclude that $T_n(f) = \frac{1}{2\pi} \int_0^{2\pi} |D_n(x)| dx$.

4 Problem 4.5.5

Question: There exists a continuous function $f(x)$ on $0 \leq x \leq 2\pi$, with $f(0) = f(2\pi)$, such that its Fourier series diverges at $x = 0$.

Proof: By (1.1) and (3.1), we know that

$$\|T_n\| \rightarrow \infty, \quad \text{as } n \rightarrow \infty,$$

which means $\{\|T_n\|\}$ is not bounded. By Banach-Steinhaus theorem, there must exist $\bar{f} \in X$ such that the set $\{T_n(\bar{f})\}$ is not bounded.

Then $s_n(0) = \frac{1}{2\pi} \int_0^{2\pi} \bar{f}(x) D_n(x) dx = T_n(\bar{f}) \rightarrow \infty$ as $n \rightarrow \infty$. That is to say, the Fourier series of $\bar{f}$ diverges at $n \rightarrow \infty$.

MA 611: Homework #5

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1 Differential operator

Question: Let $D_T = \{u(x) \in C^1[-1, 1] : \|u\|_{1,2} < \infty\}$ where

$$\|u\|_{1,2} = \left(\int_{-1}^1 u^2 dx + \int_{-1}^1 (u')^2 dx \right)^{1/2}. \quad (1.1)$$

Let $T = \frac{d}{dx}$, $D_T \subset L^2[-1, 1]$, $T : D_T \rightarrow L^2[-1, 1]$.

(a) Prove that T is not bounded.

(b) Prove that $D_{\bar{T}} \subsetneq L^2[-1, 1]$.

Proof:

(a)

Choose $f_n(x) = \sin(nt)$, since

$$\begin{aligned} \|f_n(x)\|_{1,2} &= \left(\int_{-1}^1 |f_n|^2 dx + \int_{-1}^1 |f_n'|^2 dx \right)^{1/2} \\ &= \left(\int_{-1}^1 (\sin(nt))^2 dx + \int_{-1}^1 (n \cos(nt))^2 dx \right)^{1/2} < \infty, \quad \forall n \in \mathbb{N}. \end{aligned}$$

we know that $f_n(x) \in D_T$, for each n .

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When we consider the operator $T : D_T \rightarrow L^2[-1, 1]$, the norm in D_T should be L^2 -norm.

$$\begin{aligned}
\|T\|^2 &\geq \frac{\|T(f_n)\|_{L^2}^2}{\|f_n\|_{L^2}^2} \\
&= \frac{\int_{-1}^1 |f'_n(x)|^2 dx}{\int_{-1}^1 |f_n(x)|^2 dx} \\
&\geq \frac{\int_{-1}^1 (n \cos(nt))^2 dx}{\int_{-1}^1 (\sin(nt))^2 dx} \\
&\geq \frac{n^2 \int_{-1}^1 \frac{(1+\cos(2nt))}{2} dx}{\int_{-1}^1 1 dx} \\
&\geq \frac{n^2 + \frac{1}{2}n \sin(2n)}{2} \rightarrow \infty, \quad \text{as } n \rightarrow \infty.
\end{aligned}$$

It implies that T can not be bounded.

(b) If $D_{\bar{T}} = L^2[-1, 1]$, then T should be bounded, which contradicts with the conclusion in (a).

Actually, $D_{\bar{T}} = \overline{\hat{C}^m[-1, 1]} = H^1[-1, 1] \subsetneq L^2[-1, 1]$.

2 Problem 4.6.5

Question: Let X and Y be Banach spaces and let T be a bounded linear map from X into Y . If $T(X)$ is of the second category (in Y), then $T(X) = Y$.

Proof: Since $T(X)$ is of the second category in Y , without of generality, we can assume that

$$\exists \delta > 0, \quad \text{such that } B(0, \delta) \subset T(B(0, 1)). \quad (2.1)$$

It is obvious that $0_Y \in T(X)$. Let $0 \neq y \in Y$, for $\forall \delta_1 \in (0, \delta)$, since $\frac{\delta_1 y}{\|y\|} \in B(0, \delta)$, by (2.1), we know that

$$\begin{aligned}
&\frac{\delta_1 y}{\|y\|} \in T(B(0, 1)), \\
\Rightarrow \quad &\exists x \in B(0, 1), \quad \text{such that } Tx = \frac{\delta_1 y}{\|y\|}, \\
\Rightarrow \quad &y = T\left(\frac{\|y\|}{\delta_1} x\right), \\
\Rightarrow \quad &y \in T(X).
\end{aligned}$$

Thus, $T(X) = Y$.

3 Problem 4.6.6

Question: Let X and Y be Banach spaces and let T be a linear map from a linear subspace D_T of X into Y . If D_T in X and the graph of T in $X \times Y$ are closed, then T is bounded, that is

$$\|Tx\| \leq K\|x\|, \quad \text{for all } x \in D_T, \quad K \text{ constant.}$$

Proof: Since D_T is a linear closed subspace in X and X is a Banach space, we know that D_T is also a Banach space.

By the Closed-Graph Theorem (Theorem 4.6.4), if T is a closed linear operator from a Banach space D_T to another Banach space Y , then T should be bounded.

4 Problem 4.6.7

Question: Let X be a normed linear space with any one of two norms, $\|\cdot\|_1, \|\cdot\|_2$. If $\|x_n\|_2 \rightarrow 0$ implies $\|x_n\|_1 \rightarrow 0$, then there is a constant K such that

$$\|x\|_1 \leq K\|x\|_2, \quad \forall x \in X. \quad (4.1)$$

Proof: Assume that (4.1) is not true, we want to get a contradiction.

Assume that for each n , there exists $x_n \in X$ such that

$$\begin{aligned} \|x_n\|_1 &\geq n\|x_n\|_2, \\ \Rightarrow \quad \frac{1}{n} &\geq \frac{\|x_n\|_2}{\|x_n\|_1}. \end{aligned}$$

Let $y_n = \frac{x_n}{\|x_n\|_1} \in X$, then we have

$$\|y_n\|_2 \rightarrow 0, \quad \text{but } \|y_n\|_1 = 1. \quad (4.2)$$

It contradicts with the fact that " $\|x_n\|_2 \rightarrow 0$ implies $\|x_n\|_1 \rightarrow 0$ ".

MA 611: Homework #6

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1 Problem 4.10.1

Question: A sequence $\{x_n\}$ can not have two distinct weak limits.

Proof: Suppose $\{x_n\}$ have two weak limits, x and y , then

$$\begin{aligned} x^*(x) &= \lim_{n \rightarrow \infty} x^*(x_n) = x^*(y) \\ \Rightarrow x^*(x - y) &= 0, \quad \forall x^* \in X^*. \end{aligned}$$

By Hahn-Banach theorem, we know that $x - y \equiv 0$. It implies that the weak limit should be unique.

2 Problem 4.10.2

Question: Let X be a normed linear space and let B be a dense subset of X^* . If a sequence $\{x_n\}$ in X is bounded, and if $\lim_{n \rightarrow \infty} x^*(x_n)$ exists for each $x^* \in B$, then $\lim_{n \rightarrow \infty} x^*(x_n)$ exists for all $x^* \in X^*$.

Proof: That $\{x_n\}$ is bounded means $\exists K > 0$, such that

$$\|x_n\| \leq K, \quad \forall n.$$

That B is dense in X^* means $\forall f \in X^*, \forall \varepsilon > 0, \exists f_\varepsilon \in B$ such that

$$\|f - f_\varepsilon\| < \varepsilon.$$

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That $\lim_{n \rightarrow \infty} f_\varepsilon(x_n)$ exists means $\exists N$ such that $\forall n, m > N$,

$$|f_\varepsilon(x_n) - f_\varepsilon(x_m)| < \varepsilon.$$

Thus, we have

$$\begin{aligned} & |f(x_n) - f(x_m)| \\ \leq & |f(x_n) - f_\varepsilon(x_n)| + |f_\varepsilon(x_n) - f_\varepsilon(x_m)| + |f_\varepsilon(x_m) - f(x_m)| \\ \leq & \|f - f_\varepsilon\| \|x_n\| + |f_\varepsilon(x_n) - f_\varepsilon(x_m)| + \|f_\varepsilon - f\| \|x_m\| \\ \leq & (2K + 1)\varepsilon. \end{aligned}$$

It implies that $\lim_{n \rightarrow \infty} f(x_n)$ exists for all f in X^* .

MA 611: Homework #7

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1 Problem 4.10.5

Question: A sequentially compact set is weakly sequentially compact, but not conversely.

Proof: Suppose A be a sequentially compact set, then any sequence $\{x_n\} \subset A$ has a subsequence $\{x_{n_k}\}$ that (strongly) converges to a point $x \in A$. Then

$$|f(x_{n_k}) - f(x)| = |f(x_{n_k} - x)| \leq \|f\| \|x_{n_k} - x\| \rightarrow 0, \quad \forall f \in A^*.$$

It implies that $\{x_{n_k}\}$ is weakly convergent.

However, a weakly sequentially compact set may not be sequentially, particularly in an infinite-dimensional reflexive space. Consider the unit ball in l^2 space.

Let B be the closed unit ball in l^2 . and $x_n = \{\xi_{n1}, \xi_{n2}, \dots, \xi_{nj}, \dots\} \in B$, then

$$\begin{aligned} \|x_n\|^2 &= \sum_{j=1}^{\infty} |\xi_{nj}|^2 \leq 1 \\ \Rightarrow |\xi_{nj}|^2 &\leq 1, \quad n, j = 1, 2, \dots \end{aligned}$$

We can choose a subsequence $x_{n_k} = \{\xi_{n_k1}, \xi_{n_k2}, \dots, \xi_{n_kj}, \dots\}$ such that

$$\lim_{k \rightarrow \infty} \xi_{n_kj} = \xi_j, \quad j = 1, 2, \dots$$

For any $N > 0$, we have

$$\sum_{j=1}^N |\xi_{n_kj}|^2 \leq \|x_{n_k}\|^2 \leq 1.$$

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Let $k \rightarrow \infty$, we have

$$\begin{aligned}
 & \sum_{j=1}^N |\xi_j|^2 \leq 1 \\
 \Rightarrow & \sum_{j=1}^{\infty} |\xi_j|^2 \leq 1 \\
 \Rightarrow & x_0 = \{\xi_1, \xi_2, \dots, \xi_j, \dots\} \in B.
 \end{aligned}$$

Since $(l^2)^* = l^2$, for any $y = \{\eta_1, \eta_2, \dots, \eta_j, \dots\} \in (l^2)^*$,

$$\begin{aligned}
 & |y(x_{n_k}) - y(x_0)| \\
 = & |(x_{n_k} - x_0, y)| \\
 = & \left| \sum_{j=1}^{\infty} (\xi_{n_k j} - \xi_j) \eta_j \right| \\
 \leq & \left| \sum_{j=1}^N (\xi_{n_k j} - \xi_j) \eta_j \right| + \left| \sum_{j=N+1}^{\infty} \xi_{n_k j} \eta_j \right| + \left| \sum_{j=N+1}^{\infty} \xi_j \eta_j \right| \\
 \leq & \sum_{j=1}^N |\xi_{n_k j} - \xi_j| |\eta_j| + \left(\left(\sum_{j=N+1}^{\infty} |\xi_{n_k j}|^2 \right)^{1/2} + \left(\sum_{j=N+1}^{\infty} |\xi_j|^2 \right)^{1/2} \right) \left(\sum_{j=N+1}^{\infty} |\eta_j|^2 \right)^{1/2} \\
 \leq & \sum_{j=1}^N |\xi_{n_k j} - \xi_j| |\eta_j| + 2 \left(\sum_{j=N+1}^{\infty} |\eta_j|^2 \right)^{1/2} \rightarrow 0, \text{ as } j \rightarrow \infty \text{ and } N \text{ sufficient large.}
 \end{aligned}$$

So $\{x_{n_k}\}$ is weakly convergent.

But for the sequence $\{e_n\}_{n=1}^{\infty} \subset B$ where $e_n = (0, \dots, 0, 1, 0, \dots)$, 1 in the n th place, there is no any strongly convergent subsequence.

2 Problem 4.10.9

Question: Let X and Y be Banach spaces and let $\{T_\alpha\}$ be a family of bounded linear maps from X to Y . If $\sup_\alpha |y^*(T_\alpha x)| < \infty$ for any $x \in X$, then $\sup_\alpha \|T_\alpha\| < \infty$.

Proof: For any fixed $y^* \in Y^*$, $|y^*(T_\alpha x)|$ is bounded. Then, on one hand, if we view $T_\alpha x$ as an element in the space Y^{**} and define $T_\alpha x(y^*) = y^*(T_\alpha x)$, by the principle of uniform boundedness, we know that

$$\sup_\alpha \|T_\alpha x\|_{Y^{**}} < \infty. \quad (2.1)$$

Furthermore, since $\|T_\alpha x\|_{Y^{**}} = \|T_\alpha x\|_Y$, we have

$$\sup_\alpha \|T_\alpha x\|_Y < \infty. \quad (2.2)$$

On the other hand, by (2.2), we know that $\{T_\alpha\}$ are pointwise bounded in the space $\mathcal{B}(X, Y)$, so by the principle of uniform boundedness, we can know that the set $\{\|T_\alpha\|\}$ is bounded.

3 Problem 4.10.10

Question: Let X be a normed linear space and let $\{x_n\} \subset X$. If $\{x_n\}$ converges weakly to y , then there exists a sequence $\{\sum_{i=1}^{m_n} \lambda_{i,n} x_i\}$ that converges to y .

Proof: By the theorem 4.10.7(b), since y is the weak limit of $\{x_n\}$, then $y \in \bar{Y}$ where $Y = \text{span}\{x_1, x_2, \dots, x_n, \dots\}$.

It implies that there exists a sequence $\{\sum_{i=1}^{m_n} \lambda_{i,n} x_i\}$ that converges to y .

4 Problem 4.10.11

Question: Let T be a bounded linear operator from a normed linear space X to a normed linear space Y . If $\{x_n\}$ is a sequence in X that is weakly converges to x_0 , then $\{Tx_n\}$ (in Y) is weakly convergent to Tx_0 .

Proof: Construct a operator $g = y^* \circ T : X \rightarrow Y \rightarrow \mathbb{R}$. Since $T \in \mathcal{B}(X, Y)$ and $y^* \in Y^*$, it is easy to know that $g \in X^*$.

Assume that $\{Tx_n\}$ (in Y) is **not** weakly convergent to Tx_0 , we want to get a contradiction.

If there exists $y^* \in Y^*$ such that

$$y^*(Tx_n) \nrightarrow y^*(Tx_0),$$

then there should also exist a operator $g = y^* \circ T \in X^*$ such that

$$g(x_n) \nrightarrow g(x_0),$$

which contradicts to the fact that $x_n \rightharpoonup x_0$.

Thus, $T(x_n) \rightharpoonup T(x_0)$ in Y .

5 Problem 4.10.12

Question: A sequence $\{x_n\}$ in a normed linear space X is weakly convergent to x_0 if and only if the following conditions holds:

- (i) the sequence $\{\|x_n\|\}$ is bounded,
- (ii) $x^*(x_n) \rightarrow x^*(x_0)$ as $n \rightarrow \infty$, for any x^* in a set B dense in X^* .

Proof: $\Rightarrow$:

It is obvious by the theorem 4.10.7.

$\Leftarrow$:

By (i), we know that $\exists K > 0$ such that

$$\|x_n\| \leq K, \quad n = 0, 1, \dots \quad (5.1)$$

Since B is dense in X^* , we know that $\forall x^* \in X^*, \forall \varepsilon > 0, \exists x_\varepsilon^* \in B$ such that

$$\|x^* - x_\varepsilon^*\| < \varepsilon. \quad (5.2)$$

Besides, by (ii), we know that $\exists N > 0$ such that for $\forall n > N$,

$$|x_\varepsilon^*(x_n) - x_\varepsilon^*(x_0)| < \varepsilon. \quad (5.3)$$

By Eqs.(5.1)-(5.3), we have

$$\begin{aligned} & |x^*(x_n) - x^*(x_0)| \\ \leq & |x^*(x_n) - x_\varepsilon^*(x_n)| + |x_\varepsilon^*(x_n) - x_\varepsilon^*(x_0)| + |x_\varepsilon^*(x_0) - x^*(x_0)| \\ \leq & \|x^* - x_\varepsilon^*\| \|x_n\| + |x_\varepsilon^*(x_n) - x_\varepsilon^*(x_0)| + \|x^* - x_\varepsilon^*\| \|x_0\| \\ \leq & (2K + 1)\varepsilon, \quad \forall n > N. \end{aligned}$$

So $x_n \rightharpoonup x_0$.

6 Problem 4.10.13

Question: Let K be a weakly closed set in a reflexive Banach space X . Then there exists an element $\bar{x} \in K$ such that

$$\inf_{x \in K} \|x\| = \|\bar{x}\|.$$

Thus the function $f(x) = \|x\|$ takes a minimum on K .

Proof: Let $m = \inf_{x \in K} \|x\|$. Then $\exists \{x_n\}_{n=1}^\infty \subset K$ such that

$$\lim_{n \rightarrow \infty} \|x_n\| = m.$$

By Theorem 4.10.9, $\exists$ a subsequence $\{x_{n_j}\}_{j=1}^\infty$ that converges weakly to some element $\bar{x}$. Since K is weakly closed, $\bar{x} \in K$.

Let $x^* \in X^*$, $\|x^*\| \leq 1$.

$$\begin{aligned} x^*(\bar{x}) &= \lim_{j \rightarrow \infty} x^*(x_{n_j}) \\ \Rightarrow |x^*(\bar{x})| &\leq \lim_{j \rightarrow \infty} \|x^*\| \|x_{n_j} - \bar{x}\| \leq m. \end{aligned}$$

Since X is a reflexive Banach space, we can view $\bar{x}$ as an element in X^{**} , then

$$\|\bar{x}\| = \sup_{\|x^*\| \leq 1} |x^*(\bar{x})| \leq m.$$

But on the other hand, by the definition of m , $\|\bar{x}\| \geq m$. Thus,

$$\inf_{x \in K} \|x\| = \|\bar{x}\|.$$

7 Problem 4.10.14

Question: (a) Prove that the unit ball K of a normed linear space X is weakly closed.

(b) Prove also that if the norm is strictly convex, then the minimum of $f(x) = \|x\|$ on K is attained at one point only.

Definition 7.1. A norm $\|\cdot\|$ is said to be strictly convex if $\|x\| = 1$, $\|y\| = 1$, $\|x+y\| = 2$ imply that $x = y$.

Proof:

(a) Let the sequence $\{x_n\} \in K$ and $x_n \rightharpoonup x_0$, then by Theorem 4.10.7,

$$\|x_0\| \leq \varliminf_{n \rightarrow \infty} \|x_n\| \leq 1.$$

So $\|x_0\| \in K$. It implies that any weakly convergent sequence in K has its weak limit also in K , which means K is weakly closed.

(b) Let $m = \inf_{x \in K} \|x\|$. According to Section 6, there exists at least one point $x_1 \in K$ such that $\|x_1\| = m$. If there were another element $x_2 \in K$ such that $\|x_2\| = m$, we need to prove that $x_2 \equiv x_1$.

Let $x_0 = \frac{x_1 + x_2}{2} \in K$, then

$$\begin{aligned} m &\leq \|x_0\| \leq \frac{\|x_1\|}{2} + \frac{\|x_2\|}{2} = m, \\ \Rightarrow \quad \|x_0\| &= m, \\ \Rightarrow \quad \|x_1 + x_2\| &= \|2x_0\| = 2\|x_0\| = 2m. \end{aligned} \tag{7.1}$$

Without loss of generality, we can assume that $m > 0$. By (7.1), we can get

$$\left\| \frac{x_1}{m} + \frac{x_2}{m} \right\| = 1. \tag{7.2}$$

Besides,

$$\left\| \frac{x_1}{m} \right\| = 1, \tag{7.3}$$

$$\left\| \frac{x_2}{m} \right\| = 1. \tag{7.4}$$

By the definition 7.1 and (7.2)-(7.4), we can conclude that $x_1 = x_2$.

MA 611: Homework #8

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1 Norm of strictly convex

Definition 1.1. A norm $\| \cdot \|$ is said to be strictly convex if $\|x\| \leq 1$, $\|y\| \leq 1$, $x \neq y$ imply that $\|\frac{x+y}{2}\| < 1$.

Remark 1.1. Actually, if $\|x\| < 1$, $\|y\| < 1$, then it is sure that $\|\frac{x+y}{2}\| < 1$. Besides, if $\|x\| = 1$, $\|y\| = 1$, then it is sure that $\|\frac{x+y}{2}\| \leq \frac{1}{2}\|x\| + \frac{1}{2}\|y\| \leq 1$.

So in order to prove a norm $\| \cdot \|$ is strictly convex, we just need to show that

$$\|\frac{x+y}{2}\| \neq 1, \text{ (or } \|\frac{x+y}{2}\| < 1), \text{ for } \|x\| = 1, \|y\| = 1.$$

1.1 (a)

Question: Give an example of a norm on $\mathbb{R}^2$ that is **not** strictly convex.

Solution: Consider the l^1 norm in the space $\mathbb{R}^2$. $\forall x = (x_1, x_2) \in \mathbb{R}^2$, the l^1 norm of x is

$$\|x\| = |x_1| + |x_2|.$$

Choose $x = (1, 0)$ and $y = (0, 1)$, then $z = \frac{x+y}{2} = (\frac{1}{2}, \frac{1}{2})$. Then we have

$$\|x\| = 1, \tag{1.1}$$

$$\|y\| = 1, \tag{1.2}$$

$$\|z\| = \left\| \frac{x+y}{2} \right\| = 1, \tag{1.3}$$

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which contradicts with the definition 1.1.

So the l^1 norm in $\mathbb{R}^2$ is not strictly convex.

1.2 (b)

Question: Prove X is strictly convex if and only if for each $f \in X'$, there exists at most one x with $\|x\| = 1$ so that $f(x) = \|f\|$.

Proof: $\Rightarrow$): Suppose X is strictly convex but $\exists f \in X'$ such that there are $x \neq y \in X$ satisfying $\|x\| = 1$, $\|y\| = 1$ and $f(x) = \|f\|$, $f(y) = \|f\|$. Then, on one hand,

$$f\left(\frac{x+y}{2}\right) = \frac{1}{2}f(x) + \frac{1}{2}f(y) = \|f\|. \quad (1.4)$$

On the other hand,

$$\left|f\left(\frac{x+y}{2}\right)\right| \leq \|f\| \left\|\frac{x+y}{2}\right\|. \quad (1.5)$$

Combining (1.4) and (1.5), we know that

$$\left\|\frac{x+y}{2}\right\| \geq 1,$$

which contradicts with the definition (1.1).

Thus, such f does not exist.

$\Leftarrow$): Assume that for each $f \in X'$, there exists at most one x with $\|x\| = 1$ so that $f(x) = \|f\|$.

Let $\|x\| = 1$, $\|y\| = 1$, $x \neq y$ and $z = \frac{x+y}{2}$. In order to prove the norm on X is strictly convex, we need to show that

$$\|z\| < 1. \quad (1.6)$$

By Corollary 4.8.4 in Friedman's book, for z , there exists $z^* \in X'$ such that

$$z^*(z) = \|z\|, \quad \|z^*\| = 1.$$

Let $z_1 = \frac{z}{\|z\|}$ and $z_2 = -\frac{z}{\|z\|}$, then it is easy to show that $\|z_1\| = \|z_2\| = 1$ and $|z^*(z_1)| = |z^*(z_2)| = 1 = \|z^*\|$.

Consider $\forall w \in X$ satisfying $\|w\| = 1$ and $w \neq z_1, w \neq z_2$. On one hand, by the assumption, we know that

$$|z^*(w)| \neq 1. \quad (1.7)$$

On the other hand,

$$1 = \|z^*\| = \sup_{\|w\|=1} |z^*(w)| = 1 \Rightarrow |z^*(w)| \leq 1. \quad (1.8)$$

By (1.7) and (1.8), we know that

$$|z^*(w)| < 1. \quad (1.9)$$

Since $x \neq y$, then $x, y \neq z_1, z_2$, by (1.9),

$$|z^*(x)| < 1, \quad |z^*(y)| < 1. \quad (1.10)$$

So we have

$$\|z\| = |z^*(z)| = |z^*(x/2) + z^*(y/2)| \leq \frac{1}{2}(|z^*(x)| + |z^*(y)|) < 1.$$

1.3 (c)

Question: Prove that X is strictly convex if and only if for each $z \in X$, and closed nonempty **convex** subset C , there exists at most one $x \in C$ such that

$$\|x - z\| = \inf_{y \in C} \|y - z\|. \quad (1.11)$$

Proof: $\Rightarrow$): Suppose X is strictly convex.

Let $m = \inf_{y \in C} \|y - z\|$ and $x_1, x_2 \in C$ such that

$$\|x_1 - z\| = \|x_2 - z\| = m.$$

Let $x_0 = \frac{x_1 + x_2}{2}$. Since C is convex, then $x_0 \in C$.

$$\begin{aligned} m &\leq \|x_0 - z\| = \frac{\|x_1 - z\|}{2} + \frac{\|x_2 - z\|}{2} = m \\ \Rightarrow \quad \|x_0 - z\| &= m. \end{aligned} \quad (1.12)$$

Without loss of generality, assume $m > 0$. Let $y_1 = \frac{x_1 - z}{m}$, $y_2 = \frac{x_2 - z}{m}$, then $\|y_1\| = 1$, $\|y_2\| = 1$. Besides, if $x_1 \neq x_2$, then $y_1 \neq y_2$. We want to get a contradiction.

On one hand, by Eq.(1.12), we know that

$$\left\| \frac{y_1 + y_2}{2} \right\| = \left\| \frac{x_0 - z}{m} \right\| = 1. \quad (1.13)$$

On the other hand, since X is strictly convex, we know that

$$\left\| \frac{y_1 + y_2}{2} \right\| < 1. \quad (1.14)$$

It is obviously that Eq.(1.13) contradicts with Eq.(1.14).

$\Leftarrow$): Assume that for each $z \in X$, and closed nonempty **convex** subset C , there exists at most one $x \in C$ such that (1.11) holds.

Choose C as the closed unit ball in X . $\forall x \neq y$, $\|x\| = 1$, $\|y\| = 1$, if $\left\| \frac{x+y}{2} \right\| = 1$, we want to get a contradiction.

Let $z = x + y$, then $\|z\| = 2$. $\forall w \in C$, $\|w\| \leq 1$, we know that

$$\|z - w\| \geq \|\|z\| - \|w\|\| \geq 1,$$

which implies

$$\inf_{w \in C} \|z - w\| \geq 1. \quad (1.15)$$

However, for $x, y \in C$,

$$\|z - x\| = \|y\| = 1, \quad (1.16)$$

$$\|z - y\| = \|x\| = 1. \quad (1.17)$$

By (1.15), (1.16) and (1.17), there exist at least two distinguished point x, y in C , such that

$$\|z - x\| = \|z - y\| = \inf_{w \in C} \|z - w\| = 1. \quad (1.18)$$

This contradicts with the assumption.

2 Weak * sequentially compactness

Question: Prove that $B_{X'}$ is weak * sequentially compact if X is separable.

Proof:

2.1 Introduce a lemma

Lemma 2.1. *Let X be a normed linear space and G be a dense subset of X . If the sequence $\{f_n\}_{n=1}^{\infty} \subset X^*$ satisfy these two conditions:*

- (i) $\{\|f_n\|\}_{n=1}^{\infty}$ is bounded;
 - (ii) $f_n(x) \rightarrow f_0(x)$, as $n \rightarrow \infty$, for any x in the set G (dense in X), where $f_0 \in X^*$;
- then $\{f_n\}_{n=1}^{\infty}$ is weakly * convergent to f_0 .

I want to use this lemma to prove this problem. And a short proof of this lemma will be given in the end.

2.2 Proof of this problem

Let $\{f_n\}_{n=1}^{\infty} \subset B_{X'}$, which means $\exists M > 0$ such that

$$\|f_n\| \leq M, \quad \forall n.$$

Since X is separable, we can find a dense subset of X , that is $\{x_k\}_{k=1}^{\infty}$.

Now we can use the so called "**Cantor-Hilbert diagonal method**."

Since $|f_n(x_1)| \leq \|f_n\| \|x_1\| \leq M \|x_1\|$, we know that the sequence $\{f_n(x_1)\}_{n=1}^{\infty}$ is bounded, so we can find a convergent subsequence from $\{f_n(x_1)\}_{n=1}^{\infty}$, which is

$$f_1^{(1)}(x_1), f_2^{(1)}(x_1), \dots, f_n^{(1)}(x_1), \dots$$

Then consider the sequence $\{f_n^{(1)}(x_2)\}_{n=1}^{\infty}$, then similarly, we can also find a convergent subsequence, which is

$$f_1^{(2)}(x_2), f_2^{(2)}(x_2), \dots, f_n^{(2)}(x_2), \dots$$

Keep doing this procedure, we can find this matrix of functionals on X ,

$$\begin{array}{c} f_1^{(1)}, f_2^{(1)}, f_3^{(1)}, \dots, f_n^{(1)}, \dots \\ f_1^{(2)}, f_2^{(2)}, f_3^{(2)}, \dots, f_n^{(2)}, \dots \\ \dots\dots\dots \\ f_1^{(n)}, f_2^{(n)}, f_3^{(n)}, \dots, f_n^{(n)}, \dots \\ \dots\dots\dots \end{array}$$

in which $\{f_n^{(j)}\}_{n=1}^\infty$ is the subsequence of $\{f_n^{(j-1)}\}_{n=1}^\infty$ and for each k , the sequence

$$f_1^{(k)}(x_k), f_2^{(k)}(x_k), \dots, f_n^{(k)}(x_k), \dots$$

is convergent.

Take the diagonal of the matrix, we can get

$$f_1^{(1)}, f_2^{(2)}, f_3^{(3)}, \dots, f_n^{(n)}, \dots$$

It is easy to know that for each k , $\{f_n^{(n)}(x_k)\}_{n=1}^\infty$ is convergent. That is to say, $\{f_n^{(n)}\}_{n=1}^\infty$ is bounded as well as convergent on a dense subset of X , so it should have a weak * limit in X^* according to the Lemma 2.1.

Thus, B_{X^*} is weak * sequentially compact.

2.3 Short proof of Lemma 2.1

By the assumption, we know that $\exists M = \sup_n \{\|f_n\|\} > 0$; for $\forall \varepsilon > 0$, $\forall x \in X$, $\exists x_0 \in G$ such that $\|x - x_0\| < \frac{\varepsilon}{4M}$; $\exists N$ such that $\forall n > N$, $|f_n(x_0) - f_0(x_0)| < \frac{\varepsilon}{2}$. So we have

$$\begin{aligned} & |f_n(x) - f_0(x)| \\ \leq & |f_n(x) - f_n(x_0)| + |f_n(x_0) - f_0(x_0)| + |f_0(x_0) - f_0(x)| \\ \leq & \|f_n\| \|x - x_0\| + |f_n(x_0) - f_0(x_0)| + \|f_0\| \|x - x_0\| \\ \leq & M \frac{\varepsilon}{4M} + \frac{\varepsilon}{2} + M \frac{\varepsilon}{4M} \\ = & \varepsilon. \end{aligned}$$

MA 611: Homework #9

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1 Ex. 4.13.1

Question: Chose a basis $\{e_1, e_2, \dots, e_n\}$ in $\mathbb{R}^n$ and choose a basis $\{e_1^*, \dots, e_n^*\}$ in $\mathbb{R}^{n*}$. To every linear operator T in $\mathcal{B}(\mathbb{R}^n, \mathbb{R}^n)$ corresponds a matrix (a_{ij}) , where $Te_j = \sum_{i=1}^n a_{ij}e_i$. Similarly, to every operator $S \in \mathcal{B}(\mathbb{R}^{n*}, \mathbb{R}^{n*})$ corresponds a matrix (a_{ij}^*) , where $Se_j^* = \sum_{i=1}^n a_{ij}^*e_i^*$. Find the relation between (a_{ij}) and (a_{ij}^*) if $S = T^*$.

Solution: Let $A = (a_{ij})_{n \times n}$, $A^* = (a_{ij}^*)_{n \times n}$.

$\forall x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$, $x = \sum_{i=1}^n x_i e_i$, then $Tx = Ax$.

$\forall y^* = (y_1, \dots, y_n)^T \in \mathbb{R}^{n*}$, $y^* = \sum_{i=1}^n y_i e_i^*$, then $Sy^* = A^*y^*$.

On one hand,

$$Sy^*(x) = T^*y^*(x) = y^*(Tx) = \langle y^*, Ax \rangle = \sum_{i=1}^n y_i \sum_{j=1}^n a_{ij}x_j = \sum_{i,j=1}^n a_{ij}x_j y_i.$$

On the other hand,

$$Sy^*(x) = A^*y^*(x) = \langle A^*y^*, x \rangle = \sum_{i=1}^n x_i \sum_{j=1}^n a_{ij}^*y_j = \sum_{i,j=1}^n a_{ij}^*x_i y_j.$$

It is easy to know that

$$a_{ij} = a_{ji}^*,$$

which means $A^* = A^T$.

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2 Ex. 4.13.3

Question: Let X and Y be Banach space and $T \in \mathcal{B}(X, Y)$. Prove

- (a) If T has a continuous inverse, then R_T is closed.
- (b) T^* is one-to-one $\Leftrightarrow$ if R_T is dense in Y .
- (c) If T maps X onto Y , then T^* has a bounded inverse with domain R_{T^*} .

Proof:

(a) Let $\{y_n\}_{n=1}^\infty \subset R_T$ be a Cauchy sequence in Y . Since Y is a Banach space, then $\exists y \in Y$ such that

$$y = \lim_{n \rightarrow \infty} y_n.$$

In order to prove that R_T is closed in Y , we need to show that $y \in R_T$. That is to say, to find $x \in X$ such that $y = Tx$.

Since $T^{-1} \in \mathcal{B}(X, Y)$, we know that $x_n = T^{-1}(y_n) \in X$, then

$$\|x_n - x_m\| = \|T^{-1}(y_n - y_m)\| \leq \|T^{-1}\| \|y_n - y_m\|,$$

which means $\{x_n\}_{n=1}^\infty$ is a Cauchy sequence in X . Since X is a Banach space, we can find $x \in X$ such that

$$x = \lim_{n \rightarrow \infty} x_n.$$

Furthermore, since $T \in \mathcal{B}(X, Y)$,

$$Tx = \lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} y_n = y.$$

We are done.

(b) $\Leftarrow$: By Theorem 4.13.5,

$$N_{T^*}^\perp = \overline{R_T} = Y,$$

which means

$$\forall y \in Y, \forall y^* \in N_{T^*}, \quad y^*(y) = 0. \quad (2.1)$$

Assume the T^* is not one-to-one, then $\exists y_0^* \neq 0$ such that $y_0^* \in N_{T^*}$.

So $\exists y \in Y$ such that $y_0^*(y) \neq 0$, which contradicts with (2.1).

$\Rightarrow$): Assume $\overline{R_T} \neq Y$, then by Theorem 4.13.5, $N_{T^*}^\perp \neq Y$.

Then $\exists y_0 \in Y$ such that $y_0 \notin N_{T^*}^\perp$. So $\exists y_0^* \in N_{T^*}$ such that $y_0^*(y_0) \neq 0$. It implies that $N_{T^*} \neq \{0\}$, which means T^* is not one-to-one.

(c) If $R_T = Y$, by (b), we know that T^* is one-to-one, then it has inverse $(T^*)^{-1}$ with domain R_{T^*} . It is easy to know that $(T^*)^{-1}$ is also bounded.

3 Ex. 4.13.4

Question: Let X and Y be normed linear space and let $T \in \mathcal{B}(X, Y)$. Denote $d(x, L)$ be the distance (in X) from x to a set L . Prove that

$$d(x, N_T) = \max\{x^*(x) : \|x^*\| \leq 1 \text{ and } x^* \in N_T^\perp\}. \quad (3.1)$$

Proof: On one hand, $\forall y \in N_T, \forall x^* \in N_T^\perp$ and $\|x^*\| \leq 1$,

$$\begin{aligned} x^*(x) &= x^*(x) - x^*(y) = x^*(x - y) \leq \|x^*\| \|x - y\| \leq \|x - y\|, \\ \Rightarrow \max\{x^*(x) : \|x^*\| \leq 1 \text{ and } x^* \in N_T^\perp\} &\leq d(x, N_T). \end{aligned} \quad (3.2)$$

On the other hand, by the Theorem, 4.8.3, for given $x \in X$, we can find a x^* such that

$$x^*(x) = d(x, N_T), \quad \|x^*\| = 1, \text{ and } x^*|_{N_T} = 0. \quad (3.3)$$

By (3.2) and (3.3), we can get (3.1).

4 Ex. 4.13.5

Question: Let X and Y be Banach space and $T \in \mathcal{B}(X, Y)$. If

$$R_{T^*} = N_T^\perp \quad (4.1)$$

holds, then R_T is closed.

Proof: Step One: Claim that $\exists K > 0$ such that

$$d(x, N_T) \leq K \|Tx\|. \quad (4.2)$$

For $\forall x^* \in N_T^\perp$ and $\|x^*\| \leq 1$, by (4.1), we know that there exists a y^* and $K > 0$ such that $T^*y^* = x^*$, and

$$\|y^*\| \leq K \|x^*\| \leq K.$$

So for $\forall x \in X$,

$$\begin{aligned} x^*(x) &= T^*y^*(x) = y^*(Tx) \leq \|y^*\|\|Tx\| \leq K\|Tx\| \\ \Rightarrow d(x, N_T) &\leq K\|Tx\|. \end{aligned}$$

Step Two: By (4.2), we can show that R_T is closed.

Let $\{y_n\}_{n=1}^\infty \subset R_T$ be a Cauchy sequence in Y . Since Y is a Banach space, then $\exists y \in Y$ such that

$$y = \lim_{n \rightarrow \infty} y_n.$$

In order to prove that R_T is closed in Y , we need to show that $y \in R_T$. That is to say, to find $x \in X$ such that $y = Tx$.

For each $y_n \in R_T$, we can find $x_n \in X$ such that $Tx_n = y_n$. By (4.2), we know that

$$d(x_m - x_n, N_T) \leq K\|Tx_m - Tx_n\| = K\|y_m - y_n\|.$$

Since $\{y_n\}_{n=1}^\infty$ is a Cauchy sequence in Y , we know that $\{x_n\}_{n=1}^\infty$ has a convergent subsequence in X ; without loss of generality, we also denote this subsequence as $\{x_n\}_{n=1}^\infty$. Besides, X is a Banach space, we can find $x \in X$ such that

$$x = \lim_{n \rightarrow \infty} x_n.$$

Furthermore, since $T \in \mathcal{B}(X, Y)$, we can get

$$Tx = \lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} y_n = y.$$

It implies that R_T is closed.

MA 611: Homework #10

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1 Ex. 5.1.3

Question: Let (X, μ) and (Y, ν) be measure spaces and consider the operator A defined by:

$$(Af)(x) = \int K(x, y)f(y)d\nu(y) \quad (1.1)$$

from $L^r(Y, \nu)$ into $L^p(X, \mu)$ where $1/p + 1/r = 1, 1 < p < \infty$.

Assume that X and Y are compact metric spaces, that open sets are measurable in X and Y , that μ, ν are finite measures and that $K(x, y)$ is continuous on $X \times Y$. Prove that A is compact linear operator from $L^r(Y, \nu)$ into $L^p(X, \mu)$.

Proof: Let B_1 be the unit ball in the space $L^r(Y, \nu)$. We just need to show that $\overline{A(B_1)}$ is compact.

$\forall f \in B_1, \|f\|_{L^r} \leq 1, Af \in A(B_1)$.

By assumption, $\exists M_1, M_2 > 0$ such that

$$M_1 = \max_{x, y \in X \times Y} |K(x, y)|, \quad M_2 = \mu(X)\nu(Y).$$

So we have

$$\begin{aligned} |Af(x)| &\leq \max_{x, y \in X \times Y} |K(x, y)| \mu(X)\nu(Y) \|f\|_{L^r}. \\ \Rightarrow |Af(x)| &\leq M_1 M_2 \end{aligned} \quad (1.2)$$

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For $\forall \varepsilon > 0$, $\exists \delta > 0$, such that $\forall y \in Y$, $|x - x'| < \delta$,

$$\begin{aligned} & |K(x, y) - K(x', y)| < \varepsilon, \\ \Rightarrow & |Af(x) - Af(x')| \leq \int \|K(x, y) - K(x', y)\| |f(y)| dy < \varepsilon M_2 \|f\|_{L^r}. \end{aligned} \quad (1.3)$$

Now we know that the family of functions $\{Af\} : f \in B_1$ is uniformly bounded (1.2) and equicontinuous (1.3) in $L^p(X, \mu)$, so by Arzela-Ascoli Theorem (Theorem 3.6.4), any sequence in this family has a subsequence that is uniformly convergent. It implies that $\overline{A(B_1)}$ is compact.

2 Ex. 5.1.4

Question: Let Ω_1, Ω_2 , be bounded closed sets in R^n and $X = \Omega_1$, $Y = \Omega_2$. Denote by μ the Lebesgue measure. If $K(x, y) \in L^p(X \times Y, \mu \times \mu)$, then A , defined in (1.1), is compact.

Proof: Let A be the operator defined in (1.1) where $K(x, y) \in L^p(X \times Y, \mu \times \mu)$. Choose a sequence $\{K_n(x, y)\}_{n=1}^\infty$ such that

$$\|K_n(x, y) - K(x, y)\|_{L^p(X \times Y)} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad (2.1)$$

Define a sequence $\{A_n\}_{n=1}^\infty$ as

$$(A_n f)(x) = \int K_n(x, y) f(y) dy. \quad (2.2)$$

By the conclusion of last problem, we know that each A_n is compact.

Besides,

$$\begin{aligned} & |A_n f - A f| = \left| \int (K_n(x, y) - K(x, y)) f(y) dy \right| \leq \|K - K_n\|_{L^p(Y)} \|f(y)\|_{L^r}, \\ \Rightarrow & \|A_n - A\|_{L^p} \leq \|K_n - K\|_{L^p(X \times Y)} \|f\|_{L^r}, \\ \Rightarrow & \|A_n - A\| = \sup \frac{\|A_n - A f\|_{L^p}}{\|f\|_{L^r}} \leq \|K_n - K\|_{L^p} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \end{aligned}$$

It implies that $\{A_n\}$ is uniformly convergent to A . By Theorem 5.1.2, we know that A is also compact.

3 Ex. 5.2.1

Question: Let X be a compact set in R^n and let μ denote the Lebesgue measure. Let $K(x, y) \in L^2(X \times Y, \mu \times \mu)$. For any given $g \in L^2(X, \mu)$, consider the equation, in $L^2(X, \mu)$,

$$f(x) = g(x) + \lambda \int K(x, y)f(y)dy. \quad (3.1)$$

Prove that if $g = 0$ implies $f = 0$, then there exists a unique solution of (3.1) for any $g \in L^2(X, \mu)$.

Proof: Step 1: Existence.

Define the operator $T : f(y) \mapsto \int K(x, y)f(y)dy$, then

$$g = (I - \lambda T)f,$$

where I is the identify operator.

It is easy to know that for $\lambda \neq 0$,

$$N_\lambda = \{f : \lambda f - Tf = 0\} = \{f : f - \lambda Tf = \frac{1}{\lambda}(\lambda f - Tf) = 0\},$$

and

$$R_\lambda = \{g : g = \lambda f - Tf\} = \{g : g = f - \lambda Tf = \frac{1}{\lambda}(\lambda f - Tf)\},$$

if $R_\lambda = L^2$.

The assumption that $g = 0$ implies $f = 0$ tells us that $N_\lambda = 0$. By Theorem 5.2.7, $R_\lambda = L^2$, which means $\forall g \in L^2$, there exists at least $f \in L^2$ such that $g = (I - \lambda T)f$.

Step 2: Uniqueness.

Suppose for $g \in L^2$, $\exists f_1, f_2 \in L^2$ such that

$$\begin{aligned} g &= (I - \lambda T)f_1 = (I - \lambda T)f_2, \\ \Rightarrow (f_1 - f_2) - \lambda T(f_1 - f_2) &= 0, \\ \Rightarrow f_1 - f_2 &= 0. \end{aligned}$$